Muhammad Fasih

Research Scholar & AR/VR Systems Developer

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Experience

Research Scholar & AR/VR Systems Developer Nov 2026 – Present
NED *Karachi, Pakistan*

- Research and development of augmented and virtual reality systems.

XR Engineer Aug 2026 – Oct 2026
AIDL (Associated Instrument Distributors), authorized WorldViz distributor in Pakistan *Karachi, Pakistan*

- Built interactive VR and AR simulations with WorldViz Vizard and Python.
- Modelled and prepared 3D assets for the simulation scenes in Blender.

Education

BS Computer Science 2022 – 2026
Dawood University of Engineering & Technology (DUET) *Karachi, Pakistan*

- Final year project: GeoExtract (below). 3rd place, FYDP poster competition, June 2026.
- Coursework: Computer Vision, Evolutionary Computation, Database Systems, Data Science, Embedded Systems.

Projects

GeoExtract Python, FastAPI, PyTorch, YOLOv11x-seg, EfficientNet-B2, Qwen2.5-VL, React Private repository

- Vision-language system for satellite images: a Qwen2.5-1.5B router sends each question to YOLOv11x-seg (buildings), EfficientNet-B2 (land use) or a LoRA-tuned Qwen2.5-VL 7B (visual Q&A).
- Trained on SpaceNet, WHU and INRIA on a RunPod A40; mAP50 of 81.2% (box) and 79.6% (mask). Served with FastAPI on Modal.

OpSecGuard API Python, FastAPI, RE2, WebSocket, Docker github.com/CS-Fasih/OpSecGuard-API

- Self-hosted service that redacts API keys, tokens and database URIs from production logs. Eight detectors on google-re2; 30,000+ lines per second at 48 ms P99.

Sentinel EDR C, POSIX, inotify, ClamAV, systemd, Electron github.com/CS-Fasih/Endpoint-Security-Quarantine-Tool-for-Linux

- Linux endpoint security daemon that watches /home and /tmp with inotify, scans new files through ClamAV over a UNIX socket and quarantines infected files with a JSON manifest.

Omni-Agent TypeScript, Node.js github.com/CS-Fasih/Omni-Agent

- Bring-your-own-key CLI agent for Groq, Gemini, Cerebras, SambaNova and OpenRouter. Switches provider when rate-limited and compacts context to fit smaller models. Published as `npx omnillm`.

TrueCost React, TypeScript, Express, PostgreSQL, SendGrid github.com/CS-Fasih/TrueCost

- Tracks prices across Amazon, Shopee, Lazada and Flipkart using Keepa, SerpApi and ScraperAPI, recommends buying now or waiting, and sends email alerts on price drops.

Skills

XR and 3D	WorldViz Vizard, Unity, Blender
Languages	Python, JavaScript, TypeScript, C, C#, SQL, Bash
AI and vision	PyTorch, Ultralytics YOLO, Hugging Face Transformers, LoRA fine-tuning, scikit-learn, pandas, NumPy
Web and backend	React, Next.js, Node.js, Express, FastAPI, .NET 8, REST, WebSocket
Data and tools	PostgreSQL, MongoDB, SQLite, Redis, Neo4j, Docker, Linux, Git, Vercel, Modal, RunPod

Certifications

Meta Back-End and Front-End Development (2023) · Open Source AI with Python and Hugging Face, Frontend Masters (2026) · Learning Docker, LinkedIn (2026) · Machine Learning with NumPy, pandas and scikit-learn, Educative (2025). 25 in total, listed on LinkedIn.

Other

Freelance work on Fiverr (cs_fasih) in AI/ML, Python and MERN development · Competitive chess, rated around 2200 · Languages: English, Urdu